PHYSICS 2 FINAL CHEAT SHEET

| $k=8.99\times10^9$ N·m²/C² • $\varepsilon_0=8.85\times10^{-12}$ C²/N·m² • $\mu_0=4\pi\times10^{-7}$ T·m/A • $e=1.602\times10^{-19}$ C • $m_e=9.11\times10^{-31}$ kg • $m_p=1.67\times10^{-27}$ kg • $c=3.00\times10^8$ m/s • $g=9.81$ m/s² • 1 eV=1.602×10^{-19} J | ▶ EXAM-FLAGGED ★ Cheat formula

★★★ FINAL EXAM — STARRED PROBLEM TYPES ★★★

⚠ "NEVER USE WEBERS" — keep magnetic flux as T·m², NOT Wb (1 Wb = 1 T·m²)

① Insulating Sphere — E inside/outside

▶ Uniform Q in sphere radius R. **Inside (r<R):** $E = kQr/R^3$
Outside (r≥R): $E = kQ/r^2$

Worked: E=760 N/C at r=R/4 (inside): $kQ(R/4)/R^3 = kQ/(4R^2) = 760 \rightarrow kQ/R^2 = 3040$.
At r=4R (outside): $E = kQ/(4R)^2 = kQ/(16R^2) = 3040/16 =$ **190 N/C**

If asked at r=R: both formulas give $E=kQ/R^2$

② Equivalent Resistance

▶ **Series:** $R_{eq} = R_1+R_2+R_3\ldots$
Parallel: $1/R_{eq} = 1/R_1+1/R_2+\ldots$
Two parallel: $R_{eq} = R_1R_2/(R_1+R_2)$

▶ **Method:** Reduce innermost groups first → simplify outward → $I_{total} = \varepsilon/R_{eq}$ → back-substitute V=IR for each branch.

▶ **Same R in parallel** (n of them): $R_{eq} = R/n$

③ RC Circuit (timing)

▶ **Charging:** $q(t) = Q_{max}[1-e^{-t/RC}]$, $Q_{max}=C\varepsilon$
$i(t) = (\varepsilon/R)\cdot e^{-t/RC}$
$V_C(t) = \varepsilon(1-e^{-t/RC})$; $V_R = \varepsilon\cdot e^{-t/RC}$

▶ **At t=0:** V_C=0, V_R=ε(MAX), i=ε/R(MAX), q=0
At t=∞: V_C=ε(MAX), V_R=0, i=0, q=Q_max
τ = RC (63% of final)

▶ **Discharging:** $q=Q_0e^{-t/RC}$; $i=(Q_0/RC)e^{-t/RC}$; $V=(Q_0/C)e^{-t/RC}$

Correction: exponent is −t/RC (not −ε/RC)

④ B from Moving Charge

▶ **$B = (\mu_0/4\pi)(qv\times\hat{r})/r^2$**
$|B| = (\mu_0/4\pi)\cdot(qv\sin\theta)/r^2$

▶ θ = angle between v and r. RHR for direction.

⑤ Long Straight Wire

▶ **$B = \mu_0I/(2\pi r)$** (outside, dist r from axis)

▶ **Direction:** RHR — thumb along I, fingers curl in B direction

▶ Inside (r<R): $B = \mu_0Ir/(2\pi R^2)$

⑥ Ampere's Law (Multiple Currents)

▶ **$\oint B\cdot dl = \mu_0\cdot I_{encl}$**

▶ **Sign rule:** Curl right hand along path direction. Thumb = + current direction.
★ **CCW path** → + out of page
★ **CW path** → + into page

▶ **I_encl** = sum with signs: $I_1(\pm) + I_2(\pm) + I_3(\pm)$

▶ **Then:** $B(2\pi r) = \mu_0I_{encl} \rightarrow B = \mu_0I_{encl}/(2\pi r)$

⑦ Toroid B-field

▶ **$B = \mu_0NI/(2\pi r)$** (inside, r = dist from CENTER axis)

▶ N = total turns. Outside: B = 0

⚠ Flux Φ_B = BA in T·m², not Wb

⑧ Faraday's Law of Induction

▶ **$\varepsilon = -d\Phi_B/dt$** $\Phi_B = BA\cos\theta$

▶ **If B(t) varies, θ const:** $\varepsilon = -A\cos\theta\cdot(dB/dt)$
If θ(t) varies, B const: $\varepsilon = BA\sin\theta\cdot(d\theta/dt)$
Both vary: $\varepsilon = -A[\cos\theta\cdot dB/dt - B\sin\theta\cdot d\theta/dt]$

▶ **Rotating coil:** $\theta=\omega t \rightarrow \varepsilon = NBA\omega\sin(\omega t)$; $\varepsilon_{max} = NBA\omega$

▶ **N loops:** multiply ε by N

⑨ Slide-Wire Generator

▶ **ε = vBL** (motional EMF)

▶ v = rod speed, B = field, L = rod length perp to v & B

▶ **Induced current:** $I = \varepsilon/R = vBL/R$
Force to push: $F = BIL = B^2L^2v/R$
Power: $P = I^2R = (vBL)^2/R$

⑩ RL Circuit

▶ **Energizing:** $i(t) = I_{max}[1 - e^{-Rt/L}]$; $I_{max}=\varepsilon/R$
Decay: $i(t) = I_0\cdot e^{-Rt/L}$

▶ **τ = L/R** | $V_L(t) = \varepsilon\cdot e^{-Rt/L}$ (energizing)

▶ **At t=0:** i=0 (inductor blocks). **t=∞:** i=ε/R

⑪ EM Waves — Full Set

▶ **$E(x,t) = E_m\sin(kx - \omega t)$**
$B(x,t) = B_m\sin(kx - \omega t)$

▶ **Amplitude ratio:** $E_m/B_m = c$
Instant ratio: $E(t)/B(t) = c$

▶ **Wave speed:** $c = 1/\sqrt{\mu_0\varepsilon_0} = 3\times10^8$ m/s
Wave number: $k = 2\pi/\lambda$ [rad/m]
Angular freq: $\omega = 2\pi/T = 2\pi f$ [rad/s]
c = fλ = ω/k

▶ **Poynting:** $S = (1/\mu_0)(E \times B)$ [W/m²]
$|S| = EB/\mu_0 = E^2/(c\mu_0) = cB^2/\mu_0$
Avg I: $I = E_m\cdot B_m/(2\mu_0) = \tfrac{1}{2}c\varepsilon_0E_m^2$

▶ **Direction:** E × B = propagation (RHR)

CH 21-24 CONDENSED (1 Q)

Coulomb & E-fields

$F = kq_1q_2/r^2$; $E = kQ/r^2$ (point)
Inf line: $E=\lambda/(2\pi\varepsilon_0r)=2k\lambda/r$
Inf sheet: $E=\sigma/(2\varepsilon_0)$; **Conductor:** $E=\sigma/\varepsilon_0$
Ring axis: $E=kqz/(z^2+a^2)^{3/2}$
★ Square ctr alt±q: $E=4\sqrt{2}\cdot kq/a^2$
★ Semicircle ½±Q: $E=4kQ/(\pi a^2)$
★ Zero-E pt (2 like q): $x=s/(1+\sqrt{q_2/q_1})$
★ Two ∥ sheets same sign: between=$|\sigma_A-\sigma_B|/(2\varepsilon_0)$; outside=$(\sigma_A+\sigma_B)/(2\varepsilon_0)$. Opp sign: swap.

Dipole

$\tau=p\times E$; $|\tau|=pE\sin\theta$
$U=-p\cdot E=-pE\cos\theta$; F_net=0 unif

Gauss

$\Phi=\oint E\cdot dA=Q_{encl}/\varepsilon_0$
Inside unif sphere: $E=kQr/R^3=\rho r/(3\varepsilon_0)$
Inside unif cyl: $E=\rho r/(2\varepsilon_0)$
★ Cavity in sphere: $E=\rho h/(3\varepsilon_0)$ UNIFORM

Potential

$V=kQ/r$; $U=qV$; $\Delta V=-\int E\cdot dl$; $E=-dV/dx$
$W_{field}=q\Delta V=-\Delta U$; $\tfrac{1}{2}mv^2=q\Delta V$ (rest)
Plates: $E=V/d$. Line: $\Delta V=2k\lambda\ln(r_0/r)$

Capacitors

$Q=CV$; $C=\varepsilon_0A/d$ (∥-plate); $\kappa\varepsilon_0A/d$ w/diel
Series: $1/C_{eq}=\Sigma1/C_i$; Par: $C_{eq}=\Sigma C_i$
$U=\tfrac{1}{2}CV^2=Q^2/(2C)=\tfrac{1}{2}QV$; $u=\tfrac{1}{2}\varepsilon_0E^2$
★ Discon (Q const): $U_{new}/U_{old}=d_{new}/d_{old}$
Connected (V const): $U_{new}/U_{old}=d_{old}/d_{new}$

CH 25 — EMF & CIRCUITS

$I=\varepsilon/(R+r)$; $V_{term}=\varepsilon-Ir$ (discharge)
$V=IR$; $P=IV=I^2R=V^2/R$; $W=\varepsilon It$
★ Aiding batt: $I=(\varepsilon_1+\varepsilon_2)/\Sigma R$; Opp: $I=|\varepsilon_1-\varepsilon_2|/\Sigma R$ (dir = bigger ε wins)

Current Density & Resistivity

$I = nqv_d\cdot A$ (n=charge/vol, v_d=drift)
$J = I/A = nqv_d$ [A/m²]
$R = \rho L/A$; ρ=resistivity [Ω·m]; σ=1/ρ
$E = \rho J$ (microscopic Ohm's)
Temp: $\rho(T)=\rho_0[1+\alpha(T-T_0)]$

CH 26 — KIRCHHOFF & RC

Loop: ΣV=0; **Junction:** $\Sigma I_{in}=\Sigma I_{out}$
R-series: $R_{eq}=\Sigma R_i$; **R-par:** $1/R_{eq}=\Sigma1/R_i$
$P_{dissipated}$ by $R = I^2R = V^2/R$

★ Find Time Algebra (RC/RL)

★ **Charging $V_C=\alpha\varepsilon$:** $t = -RC\cdot\ln(1-\alpha)$
α=½ → t=RC·ln(2)=0.693RC
★ **Decay to fraction f of I_0:** $t = -\tau\cdot\ln(f)$ (works RC and RL with τ=RC or L/R)
★ **RL energize to fraction f of I_max:** $t = -(L/R)\cdot\ln(1-f)$
★ **τ table:** 1τ→63%, 2τ→86%, 3τ→95%, 5τ→99.3%

★ Energy-Storage Cheat

Element	Energy	Density
Capacitor	$\tfrac{1}{2}CV^2=Q^2/2C$	$u=\tfrac{1}{2}\varepsilon_0E^2$
Inductor	$\tfrac{1}{2}LI^2$	$u=B^2/(2\mu_0)$
EM Wave	—	$u=\varepsilon_0E^2=B^2/\mu_0$

In EM wave: u_E = u_B (equal split)

★ Math Quick-Ref

Derivatives: $d/dt(e^{at})=ae^{at}$; $d/dt(\sin\omega t)=\omega\cos\omega t$; $d/dt(\cos\omega t)=-\omega\sin\omega t$

Vectors: $A\cdot B=AB\cos\theta$ (scalar); $A\times B=AB\sin\theta$ (vector ⊥ both, RHR)

Geometry: Circle area=πr^2; Sphere area=$4\pi r^2$, vol=$(4/3)\pi r^3$; Cylinder side=$2\pi rL$; Annulus=$\pi(R_2^2-R_1^2)$

Quadratic: $x = (-b\pm\sqrt{b^2-4ac})/(2a)$

Logs: $\ln(e^x)=x$; $\ln(ab)=\ln a+\ln b$; $\ln(a/b)=\ln a-\ln b$; ln(1)=0; ln(2)≈0.693

Trig: $\sin^2+\cos^2=1$; $\sin(2\theta)=2\sin\theta\cos\theta$; $\cos(2\theta)=\cos^2\theta-\sin^2\theta$

Common angles: sin/cos at 0°=0/1; 30°=½/(√3/2); 45°=√2/2; 60°=(√3/2)/½; 90°=1/0

CONSTANTS & CONVERSIONS

$k = 1/(4\pi\varepsilon_0) = 8.99\times10^9$ N·m²/C²
$\varepsilon_0 = 8.854\times10^{-12}$ C²/(N·m²)
$\mu_0 = 4\pi\times10^{-7} = 1.257\times10^{-6}$ T·m/A
$\mu_0/(4\pi) = 10^{-7}$ T·m/A
$c = 1/\sqrt{\mu_0\varepsilon_0} = 2.998\times10^8$ m/s
$e = 1.602\times10^{-19}$ C
$m_e = 9.109\times10^{-31}$ kg = 0.511 MeV/c²
$m_p = 1.673\times10^{-27}$ kg = 938 MeV/c²
g = 9.81 m/s²
1 eV = 1.602×10^{-19} J
1 N/C = 1 V/m
1 Wb = 1 T·m² ⚠ DON'T USE Wb
1 H = 1 V·s/A = 1 T·m²/A
1 Hz = 1/s; 1 rad ≈ 57.3°
1 G = 10^{-4} T; 1 mT = 10^{-3} T

UNITS REFERENCE

Quantity	Sym	Unit
Charge	q,Q	C = A·s
E-field	E	N/C = V/m
Potential	V	V = J/C
Energy	U	J = C·V
Capacitance	C	F = C/V
Current	I	A = C/s
Curr density	J	A/m²
Resistance	R	Ω = V/A
Resistivity	ρ	Ω·m
Power	P	W = J/s
EMF	ε	V
B-field	B	T = N/(A·m)
Mag flux ⚠	Φ_B	**T·m²** (NOT Wb)
Inductance	L	H = V·s/A
Reactance	X_L,X_C	Ω
Impedance	Z	Ω
Mag moment	μ	A·m² = J/T
Frequency	f	Hz = 1/s
Angular freq	ω	rad/s
Wavelength	λ	m
Wave number	k	rad/m
Period	T	s
Energy dens	u	J/m³
Intensity	I,S	W/m²
Rad pressure	P_rad	Pa = N/m²
Linear chg	λ	C/m
Surface chg	σ	C/m²
Volume chg	ρ	C/m³
Permittivity	ε₀	F/m
Permeability	μ₀	H/m
Dipole moment	p	C·m
Drift vel	v_d	m/s
Conductivity	σ	S/m

SI	p	n	μ	m	c	k	M	G
10^	−12	−9	−6	−3	−2	3	6	9

★ WORKED PRACTICE EXAMPLES & DECISION TREE

WE-1 ▸ RC Charging

R=10 kΩ, C=100 μF, ε=12 V. Find t for V_C=8 V.
τ = RC = **1.0 s**; $8 = 12(1-e^{-t/\tau}) \rightarrow e^{-t/\tau} = 1/3$
t = −τ ln(1/3) = **1.10 s**

WE-2 ▸ Long Wire B

I=5 A, r=2 cm.
$B = \mu_0I/(2\pi r) = (10^{-7}\cdot10)/(0.02) =$ **5×10^{-5} T**
Sanity: ≈ Earth's B (50 μT) ✓

WE-3 ▸ Mass Spec

V=1000 V, B=0.5 T, R=0.05 m.
$m/q = B^2R^2/(2V) = (0.25)(0.0025)/2000 =$ **3.13×10^{-7} kg/C**
Singly-ionized: m ≈ **5×10^{-26} kg** (~30 amu)

WE-4 ▸ Slide-Wire

B=0.4 T, L=0.3 m, v=2 m/s, R=5 Ω.
ε = BLv = **0.24 V**; I = **48 mA**
$F_{push} = B^2L^2v/R =$ **5.76 mN**; P = εI = **11.5 mW**

WE-5 ▸ Rotating Coil

N=50, A=0.01 m², B=0.2 T, ω=377 rad/s.
ε_max = NBAω = **37.7 V (peak)**
$V_{rms} = 37.7/\sqrt{2} =$ **26.7 V**

WE-6 ▸ Inductor Energize

ε=24 V, R=12 Ω, L=0.6 H. Find i at t=0.05 s.
τ = L/R = 0.05 s (one tau!); I_max = 2 A
$i(\tau) = I_{max}(1-e^{-1}) =$ **1.26 A**

WE-7 ▸ LRC Resonance + EM Wave

L=0.1 H, C=10 μF: $\omega_0 = 1/\sqrt{LC} =$ **1000 rad/s**; f_0 = **159 Hz**
EM wave E_m=100 V/m: $B_m = E/c =$ **3.33×10^{-7} T**; $I = \tfrac{1}{2}c\varepsilon_0E_m^2 =$ **13.3 W/m²**

★ COMMON PRODUCTS

$\mu_0/(4\pi) = 10^{-7}$ T·m/A | **$Z_0 = 1/(c\varepsilon_0) \approx 377$ Ω**
$1/\sqrt{2} \approx 0.707$; $e^{-1} \approx 0.368$; $\pi^2 \approx 9.87$
ln(2) ≈ 0.693; ln(10) ≈ 2.303; ln(3) ≈ 1.099

★ DECISION TREE — Pick Formula Fast

- **Charge in B-field, find r** → r = mv/(qB)
- **Two parallel wires** → $F/L = \mu_0I_1I_2/(2\pi d)$
- **Loop moves through B** → ε = BLv
- **B(t) changing, loop fixed** → ε = −A(dB/dt)
- **Loop rotating in const B** → ε = NBAω sin(ωt)
- **Inductor at t=0:** i=0 (open) | **at t=∞:** i=ε/R
- **Capacitor at t=0:** V=0 (wire) | **at t=∞:** i=0
- **AC find Z** → $\sqrt{R^2+(X_L-X_C)^2}$
- **AC resonance ω** → $1/\sqrt{LC}$
- **EM wave: E given, find B** → B = E/c
- **EM wave intensity** → $\tfrac{1}{2}c\varepsilon_0E_m^2$
- **Cap connected, d↑** → V const, U↓
- **Cap disconnected, d↑** → Q const, U↑
- **Wire R from material** → R = ρL/A
- **Power dissipated** → $I^2R = V^2/R$

CH 27 — MAGNETIC FORCE

- **$\vec F = q\vec v\times\vec B$**; $|F| = qvB\sin\theta$
- **F wire:** $\vec F = I\vec L\times\vec B$; $|F| = BIL\sin\theta$
- **$\Phi_B = \int B\cdot dA = BA\cos\theta$** [$T\cdot m^2$]

Circular motion

- **$r = mv/(qB)$**; $T = 2\pi m/(qB)$; $\omega = qB/m$
- $v = qBr/m$; $KE = q^2B^2r^2/(2m)$

★ Velocity selector / Mass spec

- ★ **Vel sel:** $v = E/B$ ($qE = qvB$ ↔ undeflected)
- ★ $V = Ed$ across plates
- ★ **Mass spec:** $v = \sqrt{2qV/m}$; **$m/q = B^2R^2/(2V)$**
- ★ **Free-fall into B:** $v = \sqrt{2gh}$; $F = qvB$

★ Magnetic Torque on Loop

- **$\vec\mu = NIA\cdot\hat n$** (mag dipole moment); $\hat n$ from RHR around I
- **$\vec\tau = \vec\mu\times\vec B$**; $|\tau| = \mu B\sin\theta = NIAB\sin\theta$
- **$U = -\vec\mu\cdot\vec B = -\mu B\cos\theta$**
- ★ Stable equilib: $\mu \parallel B$ ($\theta = 0$, U min). Unstable: antiparallel ($\theta = \pi$).

CH 28 — AMPERE'S LAW

- **$\oint B\cdot dl = \mu_0 I_{encl}$**
- **Biot–Savart:** $dB = (\mu_0/4\pi)\cdot(I\,d\vec l\times\hat r)/r^2$
- **Long wire:** $B = \mu_0 I/(2\pi r)$
- **Inside solid wire ($r<R$):** $B = \mu_0 Ir/(2\pi R^2)$
- **Solenoid:** $B = \mu_0 nI$ ($n = N/L$)
- **Toroid in:** $B = \mu_0 NI/(2\pi r)$; **Out:** $B = 0$
- **Ctr of circ loop:** $B = \mu_0 I/(2R)$
- **Moving pt charge:** $B = (\mu_0/4\pi)\cdot qv\sin\theta/r^2$
- ▶ **Multi-I sign:** CCW path → + out; CW → + in

Parallel wires

- **$F/L = \mu_0 I_1 I_2/(2\pi d)$**. Same dir→ATTRACT; opp→REPEL

★ Cheats

- ★ **Coax** (I_1 in, I_2 out same dir): $a<r<b$: $B = \mu_0 I_1/(2\pi r)$; $r>c$: $B = \mu_0(I_1+I_2)/(2\pi r)$. Opp&equal: $B=0$ outside.
- ★ **3 ∥ wires (eq d):** outer $F/L = \mu_0 I^2/(4\pi d)$; middle (sym)=0
- ★ **Square loop side a near wire** (ctr d, ∥): $F = \mu_0 I_1 I_2 a^2/[2\pi(d^2 - a^2/4)]$

CH 29 — FARADAY & INDUCTION

- **$\varepsilon = -d\Phi_B/dt$**; $\oint E\cdot dl = -d\Phi_B/dt$
- **$\Phi_B = BA\cos\theta$** ($T\cdot m^2$, not Wb!)
- **Lenz:** induced I opposes $\Delta\Phi$

★ EMF when B(t) and/or θ(t) vary

- ▶ **B(t) varies, θ const:** $\varepsilon = -A\cos\theta\cdot(dB/dt)$
- ▶ **θ(t) varies, B const:** $\varepsilon = BA\sin\theta\cdot(d\theta/dt)$
- ▶ **Both vary:** $\varepsilon = -A[\cos\theta\cdot dB/dt - B\sin\theta\cdot d\theta/dt]$
- ▶ **Rotating coil:** $\varepsilon = NBA\omega\sin(\omega t)$; $\varepsilon_{max} = NBA\omega$

Motional EMF — Slide-wire

- **$\varepsilon = BLv$**; $I = BLv/R$
- F to push: $F = B^2L^2v/R$; $P = (BLv)^2/R$
- ★ Loop entering/exiting B: $I = BLv/R$ during entry & exit (opp dirs); $I=0$ fully inside

★ Induced E from changing B

- ★ **Solenoid w/ dI/dt:** $r<R_{sol}$: $E = (\mu_0 nr/2)(dI/dt)$; $r>R_{sol}$: $E = (\mu_0 nR^2/(2r))(dI/dt)$; $r=0$: $E=0$
- ★ **Cyl w/ $B(t) = B_0 + B_1 t$** outside: $|E| = B_1 A/(2\pi r)$

Displacement current

- **$I_D = \varepsilon_0(d\Phi_E/dt)$**; $j_D = \varepsilon_0(dE/dt)$
- Ampere–Maxwell: $\oint B\cdot dl = \mu_0(I_C + I_D)$
- ★ Charging cap: $r<R$: $B = \mu_0 I_C\cdot r/(2\pi R^2)$; $r>R$: $B = \mu_0 I_C/(2\pi r)$

CH 30 — INDUCTANCE & RLC

- **$\varepsilon_L = -L(dI/dt)$**; $L = N\Phi_B/I$ [H]
- **Solenoid:** $L = \mu_0 N^2 A/\ell$; **Toroid:** $L = \mu_0 N^2 A/(2\pi r)$
- **$U = \frac{1}{2}LI^2$; $u_B = B^2/(2\mu_0)$**
- **Mutual ind:** $M = N_2\Phi_{21}/I_1$; $\varepsilon_2 = -M(dI_1/dt)$

RL Circuit | τ=L/R

- ▶ **Energizing:** $i(t) = (\varepsilon/R)[1 - e^{-Rt/L}]$
- ▶ **Decay:** $i(t) = I_0 e^{-Rt/L}$
- ▶ $t=0$: $i=0$ (inductor blocks); $t=\infty$: $i = \varepsilon/R$
- ★ Find L from decay: $L = Rt/\ln(I_0/I)$

LC Oscillation

- **$\omega_0 = 1/\sqrt{LC}$**; $f = 1/(2\pi\sqrt{LC})$
- $q(t) = Q\cos(\omega t)$; $i(t) = -Q\omega\sin(\omega t)$
- Energy: $U = Q^2/(2C) = \frac{1}{2}LI^2_{max}$

★ Damped LRC (series)

- ★ $\omega' = \sqrt{1/LC - R^2/(4L^2)}$
- ★ Amp decay: $A(t) = A_0 e^{-Rt/(2L)}$; t to fraction f: $t = -(2L/R)\ln(f)$
- ★ Critical: $R_c = 2\sqrt{L/C}$. $R<R_c$ under; $R>R_c$ over

CH 31 — AC CIRCUITS

- $V(t) = V_{max}\cos\omega t$; $I(t) = I_{max}\cos(\omega t - \varphi)$
- **$V_{rms} = V_{max}/\sqrt 2$**; $I_{rms} = I_{max}/\sqrt 2$
- **R:** in phase, $V_R = IR$
- **L:** V leads 90°, $X_L = \omega L$, $V_L = IX_L$
- **C:** V lags 90°, $X_C = 1/(\omega C)$, $V_C = IX_C$
- *"ELI the ICE man" — E leads I in L; I leads E in C*

Series LRC

- **$Z = \sqrt{R^2 + (X_L - X_C)^2}$**; $I = V/Z$
- **$\tan\varphi = (X_L - X_C)/R$**
- **Resonance:** $X_L = X_C$; $\omega_0 = 1/\sqrt{LC}$; $Z = R$
- **$P_{avg} = V_{rms}\cdot I_{rms}\cdot\cos\varphi = I^2_{rms}\cdot R$**

★ Phasor Diagram Tips

- ★ V_R along I (in phase); V_L straight up (+90°); V_C straight down (−90°). V_{source} = vector sum.
- ★ $|V_{source}|^2 = V_R^2 + (V_L - V_C)^2$

★ Transformer (ideal)

- ★ **$V_2/V_1 = N_2/N_1$** (turns ratio)
- ★ **$I_2/I_1 = N_1/N_2$** (inverse, conserves P)
- ★ $P_1 = P_2$ → $V_1 I_1 = V_2 I_2$ (ideal)
- ★ Step-up: $N_2 > N_1$ → V↑, I↓. Step-down: opposite.

CH 32 — EM WAVES

- ▶ **$E = E_m\sin(kx - \omega t)$; $B = B_m\sin(kx - \omega t)$**
- ▶ **$E_m/B_m = c$; $E(t)/B(t) = c$**
- ▶ **$c = 1/\sqrt{\mu_0\varepsilon_0} = f\lambda = \omega/k$**
- ▶ **$k = 2\pi/\lambda$; $\omega = 2\pi/T = 2\pi f$**
- ▶ **$S = (1/\mu_0)(\vec E\times\vec B)$**; $|S| = EB/\mu_0 = E^2/(c\mu_0) = cB^2/\mu_0$
- ▶ **$I = S_{avg} = E_m B_m/(2\mu_0) = \frac{1}{2}c\varepsilon_0 E_m^2 = E_m^2/(2\mu_0 c)$**
- **Rad pressure:** Absorb: $P = I/c$. Reflect: $P = 2I/c$
- *$\vec E\times\vec B$ → propagation (RHR). Spectrum (low→high f): Radio < μW < IR < Vis(400-700nm) < UV < X < γ*
- In medium: $v = c/n$ (n=index of refraction)

★ MAXWELL'S 4 EQUATIONS

1. **Gauss (E):** $\oint E\cdot dA = Q_{encl}/\varepsilon_0$
2. **Gauss (B):** $\oint B\cdot dA = 0$ (no monopoles)
3. **Faraday:** $\oint E\cdot dl = -d\Phi_B/dt$
4. **Ampere–Maxwell:** $\oint B\cdot dl = \mu_0(I_C + \varepsilon_0\, d\Phi_E/dt)$

★ RIGHT-HAND RULE GUIDE

- **$F = qv\times B$ (charge):** fingers along v, curl to B, thumb=F (flip if $q<0$)
- **B from wire:** thumb along I, fingers curl in dir of B
- **Ampere loop:** fingers along path, thumb = + I direction
- **Loop μ:** fingers curl with I, thumb = μ direction
- **EM wave:** fingers E, curl to B, thumb = propagation
- **Lenz:** Find dΦ direction, induced B opposes it, then RHR for I

★ COMMON PITFALLS

- ⚠ Φ_B in $T\cdot m^2$ (NOT Wb) per prof
- ⚠ RC charging is $1 - e^{(-t/RC)}$; discharging is $e^{(-t/RC)}$. Don't swap.
- ⚠ RL: at t=0 inductor BLOCKS (i=0). Capacitor at t=0 is a wire (V=0).
- ⚠ "Toroid" r is from CENTER (not from wire).
- ⚠ Cap disconnected → Q const, E unchanged when d changes. Connected → V const.
- ⚠ cos vs sin in Φ: cosθ when θ from normal; sinθ if from plane.
- ⚠ In LRC: $V_C + V_L + V_R \neq V_{source}$ (algebraic). Use phasor sum.
- ⚠ In Ampere: r is from current axis, not anywhere else.
- ⚠ Mass spec: V is accelerating voltage, not vel-selector V.
- ⚠ $\tau_{RC} = RC$, $\tau_{RL} = L/R$ — DIFFERENT formulas, easy to swap.
- ⚠ "Loop fully inside B" → no dΦ → I = 0 (only changing flux induces).
- ⚠ Direction of induced I: use Lenz, NOT the sign in Faraday's eq.

⦿ EXAM TRIAGE — KEYWORD → FORMULA

- **"insulating sphere E in/out"** → kQr/R^3 vs kQ/r^2
- **"infinite line/wire E"** → $\lambda/(2\pi\varepsilon_0 r)$
- **"infinite sheet E"** → $\sigma/(2\varepsilon_0)$
- **"conductor surface E"** → σ/ε_0
- **"point charge E"** → kQ/r^2
- **"point charge V"** → kQ/r
- **"ring on axis E"** → $kqz/(z^2+a^2)^{3/2}$
- **"semicircle ±Q"** → $4kQ/(\pi a^2)$
- **"square alt ±q"** → $4\sqrt2\, kq/a^2$
- **"cavity in sphere"** → $\rho h/(3\varepsilon_0)$
- **"flux thru hemisphere"** → $q/(2\varepsilon_0)$
- **"Φ pt charge full sphere"** → q/ε_0
- **"V from line of charge"** → $2k\lambda\ln(r_0/r)$
- **"work to move charge"** → $W = q\Delta V$
- **"speed gained from V"** → $v = \sqrt{2qV/m}$
- **"E between plates"** → V/d
- **"force on charge in E"** → $F = qE$
- **"dipole torque"** → $pE\sin\theta$
- **"dipole U"** → $-pE\cos\theta$
- **"cap C from area"** → $\varepsilon_0 A/d$
- **"cap energy"** → $\frac{1}{2}CV^2 = Q^2/(2C)$
- **"cap energy density"** → $\frac{1}{2}\varepsilon_0 E^2$
- **"cap with dielectric"** → $C = \kappa\varepsilon_0 A/d$
- **"cap disconn., move plates"** → Q const, $U\propto d$
- **"cap stays connected"** → V const, $U\propto 1/d$
- **"caps in series"** → $1/C_{eq} = \Sigma 1/C_i$
- **"caps in parallel"** → $C_{eq} = \Sigma C_i$
- **"two C in series"** → $C_1C_2/(C_1+C_2)$
- **"equiv resistance"** → reduce groups
- **"two R parallel"** → $R_1R_2/(R_1+R_2)$
- **"n equal R parallel"** → R/n
- **"power in resistor"** → $I^2R = V^2/R$
- **"power from battery"** → εI
- **"battery terminal V"** → $\varepsilon - Ir$
- **"two batteries opposing"** → $I = |\varepsilon_1 - \varepsilon_2|/\Sigma R$
- **"two batteries aiding"** → $I = (\varepsilon_1 + \varepsilon_2)/\Sigma R$
- **"wire R from L,A"** → $R = \rho L/A$
- **"drift velocity"** → $I = nqv_d A$
- **"current density J"** → I/A
- **"τ for RC"** → RC
- **"τ for RL"** → L/R
- **"RC voltage at time t"** → $V_C = \varepsilon(1 - e^{-t/RC})$
- **"RC current"** → $(\varepsilon/R)e^{-t/RC}$
- **"RC initial (t=0)"** → $V_C = 0$, $V_R = \varepsilon$, $i = \varepsilon/R$
- **"RC final (t=∞)"** → $V_C = \varepsilon$, $V_R = 0$, $i = 0$
- **"RC discharge V"** → $V_0 e^{-t/RC}$
- **"time to charge to half"** → $t = RC\ln 2$
- **"time to decay to 1/e"** → $t = \tau$
- **"loop entering B"** → $I = BLv/R$
- **"slide wire EMF"** → $\varepsilon = BLv$
- **"force to push rod"** → B^2L^2v/R
- **"power dissipated rod"** → $(BLv)^2/R$
- **"undeflected ion"** → $v = E/B$
- **"mass spec m/q"** → $B^2R^2/(2V)$
- **"speed after accel V"** → $\sqrt{2qV/m}$
- **"radius circ motion B"** → $r = mv/(qB)$
- **"period in B-field"** → $2\pi m/(qB)$
- **"cyclotron freq"** → $qB/(2\pi m)$
- **"force ∥ wires/length"** → $\mu_0 I_1 I_2/(2\pi d)$
- **"toroid B inside"** → $\mu_0 NI/(2\pi r)$
- **"toroid B outside"** → 0
- **"solenoid B"** → $\mu_0 nI$
- **"moving charge B"** → $(\mu_0/4\pi)qv\sin\theta/r^2$
- **"long wire B"** → $\mu_0 I/(2\pi r)$
- **"inside wire B (r<R)"** → $\mu_0 Ir/(2\pi R^2)$
- **"center loop B"** → $\mu_0 I/(2R)$
- **"on axis loop B"** → $\mu_0 IR^2/[2(R^2+z^2)^{3/2}]$
- **"torque on loop"** → $\mu\times B = NIAB\sin\theta$
- **"PE of dipole in B"** → $-\mu\cdot B$
- **"dI/dt in solenoid → E"** → $(\mu_0 nr/2)(dI/dt)$
- **"charging cap → B"** → $\mu_0 I_C\, r/(2\pi R^2)$
- **"displacement current"** → $I_D = \varepsilon_0\, d\Phi_E/dt$
- **"rotating coil ε"** → $NBA\omega\sin(\omega t)$
- **"max EMF rotating"** → $NBA\omega$
- **"flux when B(t)"** → $\varepsilon = -A\cos\theta\, dB/dt$
- **"flux when θ(t)"** → $\varepsilon = BA\sin\theta\, d\theta/dt$
- **"L of solenoid"** → $\mu_0 N^2 A/\ell$
- **"L of toroid"** → $\mu_0 N^2 A/(2\pi r)$
- **"inductor energy"** → $\frac{1}{2}LI^2$
- **"u_B (B-field density)"** → $B^2/(2\mu_0)$
- **"RL energizing"** → $(\varepsilon/R)[1 - e^{-Rt/L}]$
- **"RL decay"** → $I_0 e^{-Rt/L}$
- **"RL find L"** → $L = Rt/\ln(I_0/I)$
- **"LC freq"** → $1/(2\pi\sqrt{LC})$
- **"LC ω₀"** → $1/\sqrt{LC}$
- **"LRC critical R"** → $2\sqrt{L/C}$
- **"LRC underdamped ω"** → $\sqrt{1/LC - R^2/4L^2}$
- **"amplitude decay LRC"** → $e^{-Rt/2L}$
- **"X_L"** → ωL
- **"X_C"** → $1/(\omega C)$
- **"impedance Z"** → $\sqrt{R^2 + (X_L - X_C)^2}$
- **"resonance ω"** → $1/\sqrt{LC}$
- **"phase angle"** → $\tan\varphi = (X_L - X_C)/R$
- **"P_avg AC"** → $I_{rms}^2 R$
- **"V_rms"** → $V_{max}/\sqrt2$
- **"transformer V"** → $V_2/V_1 = N_2/N_1$
- **"transformer I"** → $I_2/I_1 = N_1/N_2$
- **"EM wave B from E"** → $B = E/c$
- **"intensity from E_m"** → $\frac{1}{2}c\varepsilon_0 E_m^2$
- **"Poynting"** → $(1/\mu_0)E\times B$
- **"|S|"** → $EB/\mu_0 = E^2/(c\mu_0)$
- **"rad pressure absorb"** → I/c
- **"rad pressure reflect"** → $2I/c$
- **"k from λ"** → $2\pi/\lambda$
- **"ω from f"** → $2\pi f$
- **"c from μ₀,ε₀"** → $1/\sqrt{\mu_0\varepsilon_0}$
- **"E and B in phase"** → both $\sin(kx - \omega t)$
- **"Maxwell ∮E·dA"** → Q_{encl}/ε_0
- **"Maxwell ∮B·dA"** → 0
- **"Maxwell ∮E·dl"** → $-d\Phi_B/dt$
- **"Maxwell ∮B·dl"** → $\mu_0(I_C + \varepsilon_0\, d\Phi_E/dt)$
- **"speed in medium"** → c/n
- **"f×λ"** → c (or v in medium)
- **"E density u_E"** → $\frac{1}{2}\varepsilon_0 E^2$